

Capability Matrix — v6.0 (W003 + W004)

Honest “Claim ≠ Complete” accounting. Every row is one of:

CAN (implemented & verified by runtime counts) ·

CANNOT (not possible with current source data / toolchain) ·

DID NOT / DEFERRED (possible but intentionally not done this wave).

CAN — implemented and verified

Capability	Evidence
Reduce unmapped elements via legend re-classification	982 → 112 (88.6 %)
CTM-resolved geometry extraction (bbox/centroid/shape)	<code>geometry.py</code> ; 7 shape classes
Pair text → components	315 pairs
Colour-classify text nodes	533 nodes
Pair dots / heat-load triangles / arrows → lines	205 / 100 / 132
Pairing distance quality (honest caveat)	752 pairs, median 25px. 10% above 355px flagged low-confidence; max 1040px (known outlier, deferred to W005 review)
Standardize text to Consolas + 4 mm tiers	533 nodes tiered
Assign 13 top-level layers (21 named sub-layers) per element	<code>layer_assignment.json</code>
Render layered SVG + PDF	QCELL 1834/14, RFCELL 591/12
Build interactive layer-toggle atlas	<code>layered_atlas_v6.html</code>
Trace flow arrows + junctions	132 arrows, 36 junctions
Parse vertical-letter nomenclature into tree	33 segments, 8 parents
Catalog components to Excel + HTML	297 components
Catalog spec-change dots by line	S 40, V 10, B 2, uncol. 153
Detect scope boundaries	5/5 (QM/QVB/vac/Jumper/QINFRA)
Detect handover diamonds (TP#NNN)	22
Emit PEMO YAML 1.2 SSOT	122 loops, 60 heat loads
Reconcile as-drawn catalog vs design nomenclature XLSX (W005)	97 design tags vs 141 as-drawn real tags; per-TYPE coverage delta
Detect orthogonal tag schemes (design circuit-seq vs as-drawn instance)	0 exact overlap — reported honestly, not forced
Flag instrument TYPES missing from catalog (W005)	10 design TYPES (FT/FV/HX/JJ/LE/LI/PV/RD/SV/V) absent

Capability	Evidence
Flag RFCELL template placeholders as non-reconcilable	24 placeholders (TTxxx/EHx11...) flagged
Apply documented PPT instrument re-allocations	2 (TT535→PZ coldest, TT525→PZ warmest)
Emit canonical merged instrument register (SSOT)	data/excel/canonical_register_v1.yaml , 238 entries

CANNOT — blocked by source data / toolchain

Capability	Reason
Map the 112 residual magenta elements to a line	No legend swatch exists for that colour family in the source SVG
Confirm true 3-D pipe routing / elevations	Source is 2-D schematic; no z-data
Resolve the single degenerate-transform coordinate outlier to a real position	Source transform is mathematically degenerate; mitigated by using viewBox bounds

DID NOT / DEFERRED — possible, deferred to a later wave

Capability	Status / target wave
Recover the U line (top-left, currently black)	DEFERRED — needs colour re-attribution pass
Bind the 77 floating arrows to source lines	DEFERRED — needs nearest-line heuristic
Populate 04G_E_RED line layer (geometry)	DEFERRED — RED currently text-only
Populate manifold COLD/WARM header layers (03A/03B)	DEFERRED — reserved placeholders
Design ↔ as-drawn tag cross-map (keyed on TYPE+circuit+position)	DEFERRED — W005 quantified the need (0 exact overlap); the engineering cross-map itself is the next step
Extend as-drawn catalog to the 10 missing design TYPES (FV/SV/FT/V/HX/J/LE/LI/PV/RD)	DEFERRED — W003/W004 category sheets covered only CV/EH/HV/LS/PT/TT
Exhaustive parse of the 65 MB QSYS instrumentation PPT	DEFERRED — cost; 2 documented re-allocations encoded from cited slides
Cross-drawing identity reconciliation (QCELL↔RFCELL)	DEFERRED — Wave W006
Temperature/pressure annotation per segment	DEFERRED — Wave W007
CI / GitHub Actions workflow	DEFERRED — only PLANNED in GIT-HUB_PR_PLAN.md
UI framework / web app	DEFERRED — static HTML only this wave